Noetic Diffusion Theory: Neural State, Dynamical Accessibility, and the Geometry of Possible Futures

A multiscale framework for rhythmically and contextually controlled neural dynamics

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Abstract Noetic Diffusion Theory (NDT) proposes that neural states relevant to cognition and consciousness are characterized not only by their current configuration, but also by the structured distribution of futures available from that configuration. NDT formalizes this property as dynamical accessibility: a control-dependent stochastic law governing how neural activity tends to move, how fluctuations spread, which nearby states remain probable, which longer-term destinations are favoured, and how the system responds to perturbation. Deterministic drift, diffusion geometry, local stability and reactivity, short-horizon predictive spread, basin commitment, and finite-amplitude resilience therefore describe non-equivalent aspects of the same evolving system. These properties may dissociate even when present measured states are similar, because rhythm, context, history, thalamocortical interactions, bodily state, and slower physiological constraints can reshape future transition structure without simply relocating the system in state space. NDT accordingly predicts that reduced state coordinates need not be sufficient statistics for future neural dynamics, and that matched present states can exhibit different future laws under different control conditions. Measurement charts are treated as operational interfaces to latent dynamics rather than as the theory itself. NDT is intended as a falsifiable framework for neural organization across wakefulness, sleep, altered states and pathology; it does not define a scalar of consciousness or explain why neural activity is accompanied by experience. Its central question is therefore not only where the brain is, but what it can still become.

Author Summary

Noetic Diffusion Theory is organized around a simple idea: a reduced chart of neural state is not, in general, a sufficient statistic for nearby futures. What matters scientifically is the control-dependent transition law — which directions the system tends to move, how fluctuations spread, which destinations remain open, and what processes change those possibilities. We keep a measurable state-space chart, but we treat it as an observation map rather than as the theory itself.

1. Scope and epistemic status

This article advances Noetic Diffusion Theory as a framework for neural dynamical organization associated with conscious states and cognition, not as a completed theory of why neural activity is accompanied by experience (Langell, 2026a; Storm et al., 2024). The scientifically relevant object is a neural state's **dynamical accessibility**: the geometry of a control-dependent law over possible futures, not merely a set of reachable points. A reduced chart location is not assumed to be a sufficient statistic for that law.

Selected mathematical components of NDT have been formalized and kernel-checked in Lean 4 under explicit finite-dimensional and nondegeneracy assumptions; the formalization supports the theory's architecture but does not validate empirical accessibility or biological mechanism (Supplementary Note S7).

That object is distinct from several things with which it is easily confused. NDT is not a consciousness scalar. It is not identical to Meta–Noetic Phase Space, which is a measurement chart. It is not identical to NeuralManifoldDynamics, which is a versioned empirical implementation of selected charts and estimators. It does not claim that one thalamic circuit, one rhythm, or one diffusion parameter uniquely implements consciousness, nor that every quantity named below currently has a validated empirical estimator.

1.1. Evidence classes

Every major statement in this manuscript is intended to be readable as one of five classes:

1. **Established external result** — supported by peer-reviewed literature or other independent evidence outside the present theory.
2. **Internal validated result** — supported by auditable companion code, tests, or frozen simulation/empirical contracts.
3. **Plausible interpretation** — compatible with present evidence but not directly demonstrated.
4. **Speculative extension** — a hypothesis, analogy, or forward model.
5. **Rejected or method-limited path** — tested and not supported under the stated assumptions.

Evidence class and inferential claim class are orthogonal. An internally validated result may remain only a claim-class-1 association, while a speculative mechanism can concern a claim-class-4 operator question without being empirically established.

> **Promotion rule**
>
> Success at one inferential level does not authorize a stronger claim at another. Correlation is not mechanism. Chart separation is not latent-state identification. Preserved latent ordering is not recovery of a local dynamical operator. Truth-known estimator recovery is not empirical biological validation.

The remainder of the article is organized by four layers: state geometry, local dynamical geometry, geometry of possible futures, and control of accessibility. MNPS belongs primarily to the first layer. Local Jacobians, diffusion tensors, reachability, destination geometry, and finite-amplitude questions belong to the latter three.

2. Introduction

Neural population activity is increasingly studied as trajectories through lower-dimensional state spaces rather than as collections of independently tuned variables (Breakspear, 2017; Cunningham & Yu, 2014; Vyas et al., 2020). This dynamical perspective has revealed structured population trajectories, metastable regimes, context-dependent recurrent computations, and state repertoires associated with cognition and consciousness (Barttfeld et al., 2015; Hancock et al., 2025; Hudson et al., 2014). Yet a coordinate or manifold description does not by itself identify the law governing future evolution: two systems occupying similar measured states may differ in local stability, stochastic spread, transition probability, destination structure, or response to perturbation (Duncker & Sahani, 2021).

Noetic Diffusion Theory begins from this distinction. It proposes that the scientifically relevant description of a neural regime includes not only its present measured location but its **dynamical accessibility** — the structured, context-dependent distribution of futures available from that state. Recovery from anesthesia can traverse a network of metastable activity states rather than a monotonic return to wakefulness (Hudson et al., 2014), and wakefulness has been associated with a richer repertoire of functional configurations than anesthesia (Barttfeld et al., 2015). That literature establishes the neighbourhood. NDT proposes a more explicit decomposition of the conditional future law.

NDT combines this accessibility perspective with a specific hypothesis about state-dependent control. Its minimal formulation describes rhythmically modulated stochastic dynamics on a structured landscape: Rhythmic Variance Control modulates exploratory variance, while Meta–Noetic Phase Space provides one operational chart in which state reconfiguration can be measured (Langell, 2026a). In the broader formulation developed here, this scalar variance schedule is one special case of accessibility control. Context and physiology may also alter deterministic flow, diffusion orientation, landscape mobility, or transition gates. The resulting architecture can be summarized as

$$\text{state} \rightarrow \text{local dynamical law} \rightarrow \text{possible futures} \rightarrow \text{state-dependent control of those futures.}$$

The central claim is that the functional significance of a neural state depends partly on this transition structure, not only on its present location.

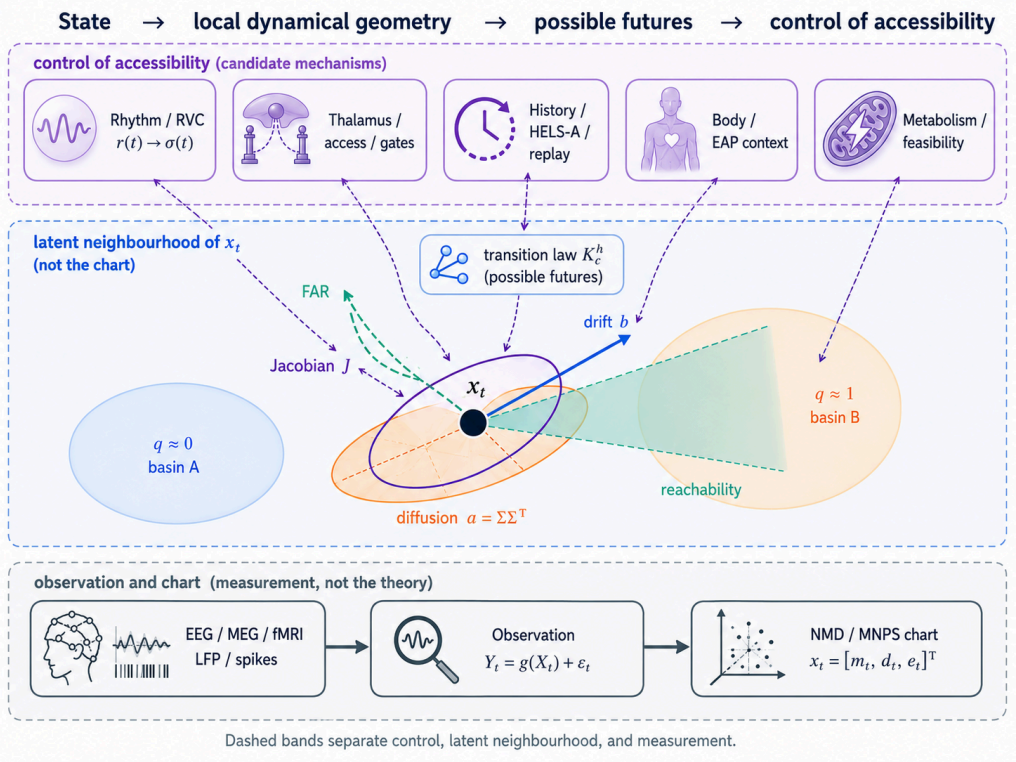


Figure 1: The NDT architecture. A measured point x_t is surrounded by local dynamical objects (drift, diffusion, Jacobian deformation, reachability, committor basins, finite-amplitude perturbation). Candidate controls act from above on different parts of that neighbourhood. EEG and related recordings enter only through an observation map and a versioned chart. Dashed bands separate control, latent dynamics, and measurement.

One state, several questions

Imagine two hikers standing at the same GPS coordinate. Their present location is identical. But one may stand on stable rock and the other on loose scree; wind may spread them in different directions; one route may lead toward one valley and another toward a second; and gates or weather may change which routes remain available.

NDT treats these as different scientific questions. State location describes where the system is. Drift describes its local tendency. Diffusion describes stochastic spread. Local deformation describes what happens to small perturbations. Predictive spread describes the distribution of nearby futures. Committors describe destination bias. Finite-amplitude resilience asks whether a larger displacement changes the regime. Context and physiology may alter these futures without necessarily changing the present measured location. A longer conceptual companion is Supplementary Note S4.

Core question

How should neural-state organization associated with cognition and conscious regimes be characterized if location and complexity are insufficient — that is, if the scientifically relevant object includes local flow, stochastic spread, reachable futures, basin commitment, and the processes that reshape them — without mistaking a measurement chart for the latent process or a correlation for a dynamical operator?

2.1. Three questions

The framework is organized around three questions that later sections develop in turn.

Where is the system? This is the measurement-chart question. A low-dimensional representation

$$x_t = [m_t, d_t, e_t]^\top \tag{1}$$

or a stratified nine-dimensional refinement, reports an operational location. It answers a **state** question.

What can happen from here? This is the dynamical core of NDT. It includes deterministic drift, stochastic diffusion, transient amplification, short-horizon reachability, basin commitment, and finite-amplitude resilience. These objects answer different dynamical questions and are not interchangeable summaries of one another.

What changes what can happen? This is the control layer: rhythms, candidate thalamic access, attention and context, history and replay, bodily state, metabolic feasibility, and — as an explicitly speculative micro substrate — slow structural accessibility at dendritic scale. Control may act on drift, diffusion, effective geometry, landscape, or transition gates, not only on a scalar noise amplitude.

2.2. Noetic accessibility

Noetic accessibility is the geometry of a control-dependent transition law over future neural states and destinations. "Noetic" is retained as the historical name of the theory and marks its intended domain of mind-relevant neural dynamics; no semantic, epistemic, or phenomenal property is inferred from the coordinates themselves (Langell, 2026a). It does not mean conscious access in the global-workspace sense (Dehaene & Changeux, 2011). Stochastic short-horizon spread is not workspace ignition. Chart-level accessibility is not a report. A conceptual companion to this architecture is Supplementary Note S4.

3. The NDT scale and inferential claim classes

3.1. Latent process, observation, and chart

Let X_t denote a latent neural dynamical state. Its exact dimensionality and biological substrate need not be known for the theory to be well posed as a modelling class. Measurements arise through an observation map

$$Y_t = g(X_t) + \varepsilon_t, \tag{2}$$

where Y_t may be spikes, local field potentials, EEG, MEG, fMRI, or physiological channels, and ε_t is observation noise. A subsequent feature and chart construction produces an operational representation

$$Y_t \to z_t \to x_t. \tag{3}$$

The chart x_t is a versioned summary of the measurement. It is not automatically the latent manifold. Different modalities, preprocessing decisions, neighbourhood definitions, and projections yield different charts, and their uncertainty is inherited by every later estimate of drift, Jacobian, diffusion, or reachability.

3.2. Four claim levels

Companion studies of observability and local-operator recovery under known ground-truth dynamics motivate four **inferential claim classes** that govern strength throughout this article. The numbering reflects increasing typical inferential burden, not a strict logical ordering:

1. **Association.** Does an observable covary with another measured or latent quantity under stated nulls?
2. **Mechanism discrimination.** Can observations distinguish alternative mechanisms that produce similar task-level outputs?
3. **Latent-order preservation.** Does the measured representation preserve a predefined ordering of a latent summary?
4. **Dynamical-operator identification.** Can the representation recover a local operator that correctly predicts responses to perturbation or short-horizon evolution?

These levels are not a general nonlinear-observability theorem. They are claim classes, and they are one of the paper's methodological contributions: an operational chart can succeed for one scientific question while failing for another. Detectable population information, chart association, local-operator recovery, and mechanism identification are not interchangeable. Related recent work likewise emphasizes that latent state-space fits can capture statistical associations without automatically identifying dynamics that predict responses to perturbation (Nejatbakhsh & Wang, 2025), and that low-dimensional manifold descriptions do not by themselves identify the dynamical rule on that manifold (Cunningham & Yu, 2014; Duncker & Sahani, 2021). The order is not monotonic: a chart can discriminate mechanisms while failing to preserve order or to identify an operator.

A companion meso-to-macro analysis illustrates the distinction: cross-scale associations survive declared nulls, whereas the corresponding EEG chart fails its local-operator estimability criterion (Claar et al., 2023; Langell, 2026b). A complementary IBL analysis similarly finds decodable population information without a valid nominated chart-transition operator (Section 10) (Langell, 2026c). Failure of a dynamical readout need not imply absence of information in the underlying population.

Consequence

MNPS separations can be meaningful at claim classes 1–3. Claims about Jacobians, diffusion tensors, reachability, committors, or finite-amplitude resilience require stronger validity conditions and are not licensed when those conditions are not met.

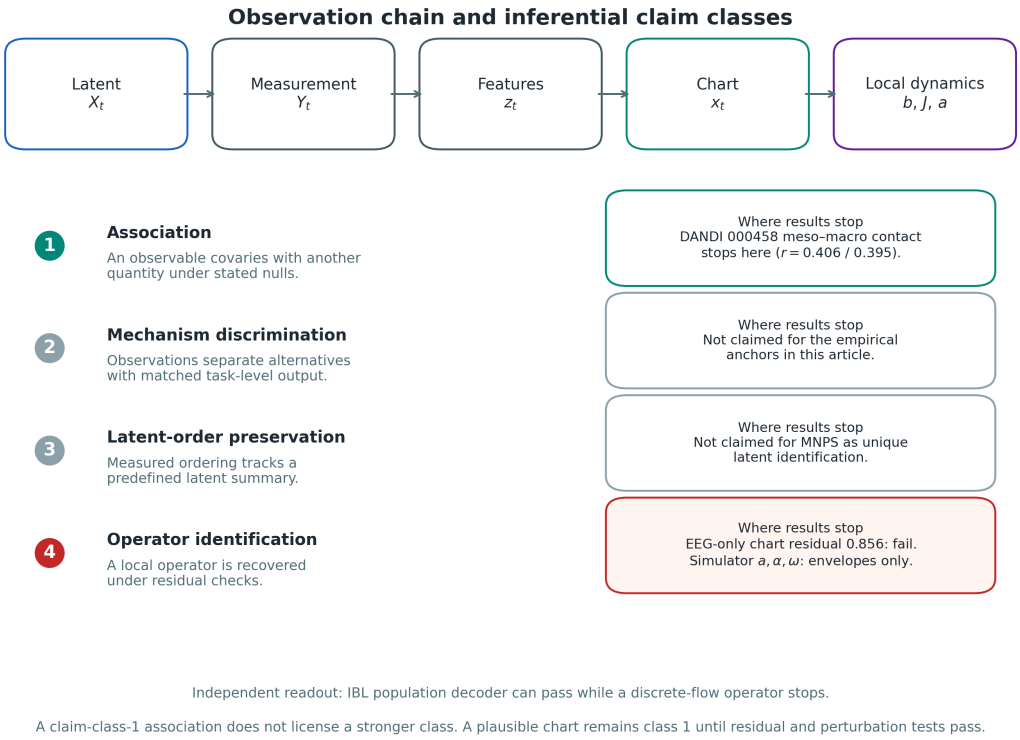


Figure 2: Observation chain and inferential claim classes. Companion meso–macro contact (DANDI 000458) is a claim-class-1 constraint; the same EEG chart fails operator identification. An independent IBL readout shows the same lesson: population decoding can succeed while a nominated chart-transition operator does not (Section 10). Recovery of a, α, and ω on systems with known ground-truth dynamics remains envelope-limited.

4. Core NDT dynamics

4.1. Canonical stochastic formulation

The working process class of the present theory is a controlled diffusion, not a pure gradient flow. Let c_t denote a contextual or control state. The latent dynamics are written

$$dX_t = b(X_t, t \mid c_t)dt + \Sigma(X_t, t \mid c_t)dW_t, \tag{4}$$

where b is local deterministic drift, Σ determines the amplitude and orientation of stochastic forcing, and W_t is Brownian motion. Equation 4 is a modelling class, not a fitted neural SDE. The diffusion formulation is an effective continuous-state approximation, not a claim that neural fluctuations are literally Brownian or that all relevant neural dynamics are Markovian. Hybrid, colored-noise, delayed, or non-Markov formulations remain compatible extensions when required by the observation scale.

4.2. Dynamical accessibility as a transition law

If X_t were a complete Markov state and the future control protocol were known, the present latent location would determine the conditional future law. That is not the empirical situation. The load-bearing claim of NDT is that a **reduced measurement chart** x_t is generally not a sufficient statistic for the latent future law K^X.

The infinitesimal generator of Eq. 4, acting on a test function f, is

$$\mathcal{L}_{t \mid c} f(\xi) = b(\xi, t \mid c) \cdot \nabla f(\xi) + \frac{1}{2} \operatorname{tr}\big(a(\xi, t \mid c) \nabla^2 f(\xi)\big), \tag{5}$$

with $a = \Sigma\Sigma^\top$ as in Eq. 8. Dummy ξ is a realization of the **latent** state X_t, not of the measurement chart x_t. Because c_t and history may vary over a horizon, the latent protocol-conditional law is

$$K^{X \mid \gamma}_{t:t+h}(\xi, A) = P\big(X_{t+h} \in A \mid X_t = \xi \text{ and } \gamma_{[t:t+h]}\big), \tag{6}$$

where $\gamma_{[t,t+h]}$ is the control/history protocol on that interval (an instantaneous c_t, a path of c, or an augmented state $Z_t = (X_t, C_t, H_t, ...)$ made approximately Markov). Experiments generally estimate a chart-conditional approximation

$$K^{x \mid \gamma}_{t:t+h}(x, B) = P\big(x_{t+h} \in B \mid x_t = x \text{ and } \gamma_{[t:t+h]}\big). \tag{7}$$

The latent process has a transition law K^X; measured work generally recovers projections or chart-conditional laws K^x. NDT-C1 states that a reduced chart coordinate need not be sufficient for K^X. When the text writes K^h without a superscript, the intended object is K^x unless a latent argument is explicit. Conditioning only on a frozen snapshot of c_t is a special case, not the general theory. History dependence (HELS-A, P6) is a statement about γ, not a violation of Eq. 4.

K^h is the law of the **state at horizon h**, not the full path-space measure on $X_{[t,t+h]}$ or $x_{[t,t+h]}$. NDT uses the family $\{K^h\}_{h>0}$ as its primary finite-horizon object. Path-dependent quantities such as committors are defined under the corresponding stochastic process (E & Vanden-Eijnden, 2010; Klus et al., 2018; Metzner et al., 2009). A reachable **set** discards probability: two systems can share support and still differ in mass.

These named summaries are different **kinds** of object, not interchangeable views of one endpoint distribution. The generator $\mathcal{L}$ is the infinitesimal law; K^h is the finite-horizon law of the state at $t + h$; committors q are path/hitting functionals; FAR is an interventional finite-amplitude response. Calling them all accessibility does not identify them.

The following are non-equivalent functionals or descriptors of the controlled process and its transition laws, not rival theories:

- b — infinitesimal conditional mean rate;
- a — infinitesimal conditional covariance rate;
- J — local drift derivative in a declared chart;
- stochastic short-horizon reachability — finite-horizon predictive spread;
- committor q — destination (hitting) probability;
- attractors — long-time organization;
- FAR — response to a finite displacement.

Rhythm, thalamus, history, body, and metabolism are candidate changes of K or $\mathcal{L}$, not extra definitions of NDT. The generator, Itô corrections, and estimator validity conditions are spelled out further in Supplementary Methods S1.

Three core commitments

NDT-C1 (reduced-state insufficiency). Current location in a reduced measurement chart x_t need not determine the latent future law K^X.
NDT-C2 (non-redundant accessibility). In at least some distinct neural regimes, functionals of the chart-conditional law provide reproducible out-of-sample information not contained in current chart location and standard static or spectral summaries.
NDT-C3 (conditional control). Context, history, or physiology can alter $K^x_h(\cdot \mid x, \gamma)$ among observations matched on current chart state.

C1 alone is not distinctive: any coarse measurement chart may fail to be sufficient for latent dynamics. The empirical content of NDT lies in C2 and C3 — whether structured future-law quantities add reproducible information beyond present-state summaries, and whether context, history, or physiology systematically reshape those futures at matched measured state.

The experimental statement of C3 is $K_h^{x \mid \gamma_1}(x, \cdot) \neq K_h^{x \mid \gamma_2}(x, \cdot)$ at matched chart x. That is not the generic observation that adding covariates improves prediction. Its empirical content is that predefined physiological or contextual variables systematically alter nominated features of the future law at matched chart state, beyond standard state, spectral, and nuisance covariates. That is the common form of rhythmic, thalamic, historical, and embodied control. Context-dependent recurrent computation is an established external precedent for such protocol-conditional dynamics; it does not test NDT (Mante et al., 2013).

C1 asks whether present chart state determines nearby futures. C2 asks whether accessibility quantities add non-redundant information. C3 asks whether matched states can have different futures under different control conditions.

4.3. Diffusion geometry

Diffusion in this theory is not a scalar noise amplitude. Define the local diffusion tensor

$$a(\xi, t) = \Sigma(\xi, t)\Sigma(\xi, t)^{\top}. \tag{8}$$

The tensor a determines the magnitude, effective dimensionality, anisotropy, and preferred directions of stochastic spread. Two states with identical mean drift may differ profoundly in what fluctuations permit. Scalar $\sigma(t)$ is the isotropic special case.

4.4. Potential-driven and non-conservative drift

Potential-driven dynamics form one structured component of the drift, not an assumption that every neural trajectory is gradient descent:

$$b(\xi, t \mid c) = -M(\xi, t \mid c)\nabla V(\xi, t \mid c) + u_{\mathrm{nc}}(\xi, t \mid c). \tag{9}$$

Here V is a state-dependent potential (not IIT's Φ) and $M \succ 0$ is a **mobility tensor**: the matrix that converts a Euclidean gradient into potential-driven drift. If a Riemannian metric g is later declared, $M = g^{-1}$. NDT does not currently assume a fluctuation–dissipation relation linking M to the diffusion tensor a. The residual u_{nc} is permitted because circulation-like, cyclic, or asymmetric flows need not be gradient descent under the declared geometry. The decomposition is not uniquely identified from trajectories: anisotropic or state-dependent mobility can itself generate non-symmetric chart Jacobians and apparent non-normality. Jacobian asymmetry is therefore not sufficient evidence for a non-conservative mechanism.

Dummy ξ remains a realization of latent X_t. Later sections discuss empirical counterparts as chart-level estimates $\hat{b}_x$, $\hat{a}_x$, and $\hat{J}_x$ on a declared measurement chart x.

4.5. Rhythmic variance control as the minimal special case

Rhythmic Variance Control is the minimal isotropic accessibility-control model (Langell, 2026a). Let $r(t) \in [0, 1]$ denote an operational rhythmic-control proxy and

$$\sigma(t) = \sigma_{\min} + \sigma_0(1 - r(t)). \tag{10}$$

High $r(t)$ therefore reduces stochastic amplitude and biases the process toward stabilization, whereas lower $r(t)$ permits broader exploration. In the general NDT process, rhythmic control need not act only on scalar variance:

$$r(t), c(t) \to \{b_t, \Sigma_t, M_t, \text{transition gates}\}. \tag{11}$$

RVC is thus the isotropic variance-control limit of the broader accessibility-control hypothesis. State- or event-dependent changes in neural variability are themselves an established cortical phenomenon (Churchland et al., 2010). NDT's stronger claim concerns structured rhythmic control of such variation, not the existence of variability quenching. Stronger specificity claims are treated as control-layer hypotheses in Section 9, not as denoising proofs.

5. Measured state: MNPS and the NeuralManifoldDynamics boundary

5.1. Canonical and stratified charts

The three-dimensional observation chart is Eq. 1. Operational families: m indexes mobility, metastability, or rhythmic coordination; d indexes deviation from an integration–segregation balance; e indexes entropy or complexity pressure. These are chart semantics, not unique biological axes.

A stratified nine-dimensional chart

$$x_t^{(9)} = [m_a, m_e, m_o, d_n, d_l, d_s, e_e, e_s, e_m]^{\top} \tag{12}$$

provides higher-resolution measurement, with recomposition into the three families through a release-fixed projection $x_t = P_{\text{fixed}} x_t^{(9)}$. The rationale is to reduce dimensional masking and to expose compensatory changes hidden in 3D. P_{fixed} is an auditable operational contract, not a claim of unique ontology. Downstream local-dynamics claims inherit that contract.

5.2. MNPS is not NDT

MNPS provides coordinates in which some NDT-related quantities can be estimated. It does not exhaust NDT. The theory remains meaningful if another valid chart replaces MNPS, if some dynamical objects are more

identifiable than the coordinates themselves, or if modality-specific charts differ. Identifying the latent state X_t with (m, d, e) is rejected as a homology. Dimensionality reduction supplies a parsimonious population description; it does not automatically identify the underlying process (Cunningham & Yu, 2014; Duncker & Sahani, 2021).

5.3. NeuralManifoldDynamics as a measurement contract

NeuralManifoldDynamics (NMD) operationalizes selected MNPS and local-dynamics quantities through versioned preprocessing, feature construction, chart projection, and estimator contracts. MNPS is an observation chart through which some NDT quantities can be estimated, not the theory itself.

NDT specifies latent interpretation, the objects b, a, J, reachability, committor, and finite-amplitude questions, predicted dissociations, and mechanism hypotheses, whereas NMD operationalizes the corresponding measurement contracts.

A Jacobian estimated specifically in MNPS is termed an MNJ; its derived quantities remain chart-level measurements. Diffusion, stochastic reachability, committor, and finite-amplitude families require independent estimator validation and are not implied merely by availability of an MNPS chart or Jacobian. Simulator recovery of diffusion tensors and Jacobian summaries on linear SDEs does not, by itself, license those quantities on empirical charts. Current software status and estimator validity conditions are documented in Supplementary Methods S1.

5.4. Measurement inheritance rule

Any downstream inference inherits the validity limits of (i) raw measurement, (ii) preprocessing, (iii) feature construction, (iv) chart definition, and (v) local estimator. A plausible embedding does not cancel those limits.

6. Local dynamical geometry

Empirical local flow is estimated on a declared measurement chart. Write $\hat{b}_x$ for a chart-level drift estimate. The empirical Jacobian is

$$\hat{J}_x(x, t) = \frac{\partial \hat{b}_x}{\partial x}. \tag{13}$$

The corresponding latent derivative is $\nabla_\xi b(\xi, t)$. Unadorned J in later empirical sentences means $\hat{J}_x$ unless a latent argument is explicit. The term MNJ is reserved for a Jacobian estimated in an MNPS chart. $\hat{J}_x$ asks how nearby perturbations change under the local chart flow. It does not ask where the system sits, how noise spreads, or which basin is favoured.

Two derived summaries are conceptually distinct and must remain so. The **spectral abscissa**

$$\alpha(J) = \max_i \Re[\lambda_i(J)] \tag{14}$$

is the dominant modal growth or decay rate. The **numerical abscissa**

$$\omega(J) = \lambda_{\max}\left(\frac{J + J^\top}{2}\right) \tag{15}$$

is the maximal instantaneous growth rate. A system can satisfy $\alpha < 0$ while $\omega > 0$: every eigenmode decays, yet some perturbations grow at short times. Non-normal transient amplification of this kind is an established property of neuronal networks, distinct from eigenvalue-driven slowing (Christodoulou & Vogels, 2022; Hennequin et al., 2012). The reactivity gap $\Delta_{\text{react}} = \omega - \alpha$ is a derived, non-normality-sensitive summary of that separation. It is not a complete measure of non-normality. The antisymmetric part $\Omega = \frac{J - J^\top}{2}$ supplies a rotational component distinct from reactivity.

Finite-time amplification, for example a peak gain $G_{\text{peak}}(T)$ of a local propagator over a horizon T, is a different object again. Companion simulator studies with known ground-truth dynamics recover α, ω, and Δ_{react} on declared constant-linear systems under restricted envelopes, and show that they are not interchangeable (Langell, 2026d). Instantaneous Jacobian recovery does not automatically license ordered finite-time summaries, and simulator success does not automatically transfer to empirical charts. Those boundaries stay in force here: "stable" is not a sufficient dynamical phenotype.

Coordinate boundary. The quantities J, α, ω, and Ω used here are local summaries in a declared measurement chart. Away from fixed points they are not generally invariant under nonlinear reparametrization, and an Itô SDE acquires an Itô correction under a nonlinear chart change. NDT therefore treats current MNJ-derived quantities as **chart-level dynamical measurements** rather than intrinsic manifold tensors. An intrinsic

formulation would require an explicitly declared geometric connection and a covariant derivative; that object is not currently estimated and is not claimed.

7. Local diffusion geometry

Equation 8 already makes diffusion a tensor. The questions it answers are not interchangeable with those answered by J. Natural summaries include total diffusion $D_{\text{total}} = \text{tr}(a)$, an effective diffusion dimensionality, a concentration or anisotropy index, and orientation-specific alignments relative to a declared probe direction.

Two states can share a similar J and differ in a, or share a similar a and differ in J. That dissociation is part of what "diffusion" is supposed to mean in NDT. Companion studies with known ground-truth dynamics recover a on linear SDEs as a mathematical measurement under inspectable envelopes (Langell, 2026d) (support, timestep, bandwidth, observation noise). Gauge matters: orthogonal rotations leave tensor summaries invariant to numerical tolerance, whereas nonuniform rescaling does not. Recovered a is therefore chart-dependent unless a chart is declared. Empirical interpretation through NMD remains a separate translation problem; diffusion geometry is not implied by Jacobian residuals alone.

8. Geometry of possible futures

Local linearization does not exhaust accessibility. This section collects objects that ask what can happen over a finite horizon, toward which destination, and after a perturbation that is no longer infinitesimal.

8.1. Stochastic short-horizon reachability

Stochastic short-horizon reachability asks how probability mass over nearby future states spreads under K^h. It is a covariance-based approximation to finite-horizon **predictive spread**, not classical control-theoretic reachability and not a Gramian of a structural connectome. A short-horizon, locally frozen-coefficient stochastic linearization about the reference state at time t uses A_t, b_t, and Q_t held fixed over the propagation horizon:

$$x_{t+1} = A_t x_t + b_t + \varepsilon_t, \quad \varepsilon_t \sim (0, Q_t) \tag{16}$$

induces the locally frozen-coefficient covariance recursion

$$\Sigma_t^{(h+1)} = A_t \Sigma_t^{(h)} A_t^\top + Q_t, \tag{17}$$

with $\Sigma_t^{(0)} = 0$. This recursion approximates short-horizon covariance propagation under the local linear model; it is not the general time-varying transition law. Eigenvalues of $\Sigma_t^{(h)}$ yield volume, anisotropy, and effective-dimension summaries of a **predictive-spread ellipsoid**. That ellipsoid is not an unqualified reachable set: it describes second-moment spread under a local linearization, not the support of every admissible control.

Predictive spread is not occupancy, entropy, traversability, controllability, or consciousness. Large Q_t can inflate the ellipsoid without richer controllable geometry. Horizon and neighbourhood leakage are validity conditions. Companion simulator work can calibrate aggregate covariance propagation while failing tests of memory-specific or target-private directional spread; calibration is not destination geometry. Empirical claims that invoke accessible futures therefore require their own estimator validation; they cannot be read off chart location.

8.2. Committor geometry

For competing destinations A and B, the committor

$$q(x) = P_x(\tau_B < \tau_A) \tag{18}$$

is the probability of hitting B before A. It is destination geometry, not local expansion and not reachable volume. The object is theoretically central and is the continuous-process counterpart of transition-path theory (E & Vanden-Eijnden, 2010; Metzner et al., 2009). Its empirical and even finite-sample status is severely bounded. One-dimensional local-law recovery can be qualified under restricted contracts; two-dimensional finite-sample recovery has not been supported under the estimator families so far tested. Current validation does not support empirical committor estimation within the NMD pipeline. Stage labels are not committor truth. Committor geometry is therefore retained as a theoretical target rather than treated as a currently validated empirical measurement.

8.3. Attractors and finite-amplitude resilience

Attractor and basin structure — locations, depth, boundaries, corridors, metastability, escape — organize longer-run trajectories (Deco et al., 2017; Hancock et al., 2025; Tognoli & Kelso, 2014). Visual clustering is not evidence of an attractor. Finite-amplitude resilience (FAR) asks how large a perturbation the system can tolerate before leaving its current dynamical regime. Local stability does not imply FAR; Jacobian or finite-time tangent

gain does not substitute for an observed perturbation protocol. Without such a protocol, FAR is not an empirical measurement.

8.4. Non-equivalent dynamical questions

The point of this hierarchy is not to brand a new metric for each row. It is to keep the questions distinct. They are non-equivalent in explanatory target, not orthogonal in the linear-algebraic or statistical sense.

Dynamical question	Object
Where is the system?	chart state x
Where is it locally moving?	drift b
Do infinitesimal modes grow?	spectral abscissa α
Can perturbations initially amplify?	numerical abscissa ω
How does random variation spread?	diffusion tensor a
How does short-horizon mass spread?	predictive-spread ellipsoid
Which destination is favoured?	committor q
Which basin organizes the trajectory?	attractor geometry
Can a finite perturbation dislodge it?	FAR

Table 1: Non-equivalent questions in NDT. No single local measure answers the others. Empirical availability of each object is a separate contract, not a consequence of having a chart.

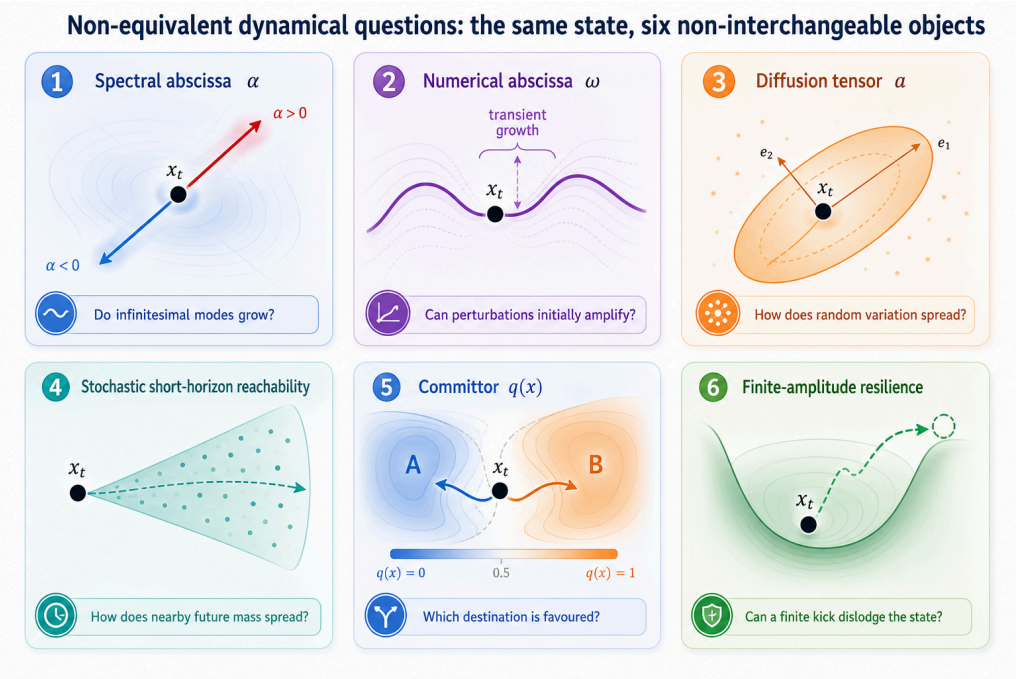


Figure 3: Non-equivalent dynamical questions. The same state x_t supports six non-interchangeable objects. No single local measure answers the others.

Formal neighbours — network control, predictive processing, metastability, and the major consciousness theories — are compared in Section 11. The present emphasis remains measured dynamical accessibility rather than control energy, free-energy identity, workspace ignition, or integrated information.

9. Biological control of accessibility

Equation 4 is incomplete until one says what c_t is allowed to change. The theory does not need a catalogue of subsystems. It needs candidate implementations of C3: processes that can change $K_h^x(\cdot \mid x, \gamma)$ at matched chart location. The main text groups candidate implementations into three broad classes. They occupy different slots in $\{b, \Sigma, M, V, \text{gates}\}$ and are not interchangeable. Cortical gain itself can be reshaped by attention, arousal, motor state, and neuromodulation through multiple cellular and circuit routes (Ferguson & Cardin, 2020); NDT's control layer is a candidate decomposition of how such influences change K_h^x, not a claim that those routes are

unique to the present theory. A failure in one slot revises that implementation; it does not falsify accessibility control as a modelling class.

Candidate	**Intended action**	**Present ceiling**
RVC	scalar $\sigma(t)$	minimal special case; event-gated write specificity not supported
Thalamic access	directional A_{sig}, A_{noise} on frozen V	synthetic operator existence; biology is association, not operator ID
HELS-A	context operator A_H on frozen V	synthetic access without landscape rewrite; empirical scoring not licensed
HELS (legacy)	$\partial V/\partial t$ from history	landscape problem remains open; infrastructure is not writing
Embodiment / metabolism	context c_t^{body}; feasibility q^{met}	conditioning and constraint; not consciousness coordinates

Table 2: Control of accessibility. Each row changes a different part of the local dynamical object. None is licensed as **the** NDT mechanism.

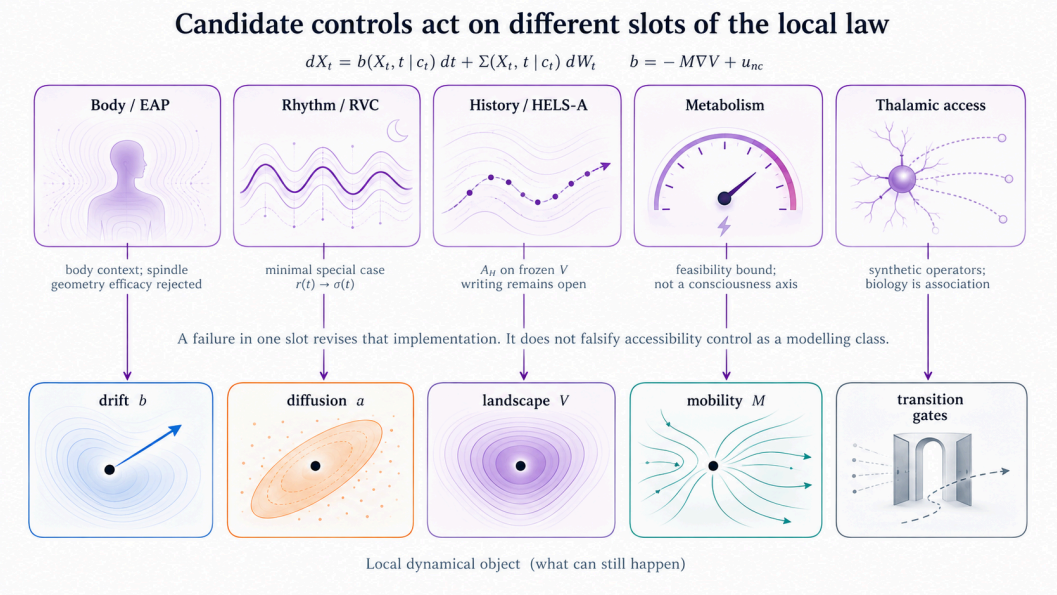


Figure 4: Accessibility control. Each candidate mechanism is drawn as acting on a distinct slot of the local law. A failure in one slot revises that implementation; it does not falsify the modelling class.

9.1. Fast rhythmic and access control

Equation 10 is the isotropic variance-control limit: a rhythmic proxy $r(t)$ schedules scalar noise amplitude and thereby the exploration–stabilization trade-off. Event- or state-dependent quenching of trial-to-trial variability is an established external phenomenon (Churchland et al., 2010). That does not establish RVC. The NDT claim is stronger and narrower: structured rhythmic control of such variation, of which Eq. 10 is the minimal special case. That is a claim about **one** entry in $\{b, \Sigma, M, V, \text{gates}\}$. Companion tests of stronger write-gate timing against energy-matched and shuffled-timing controls have not supported a general rhythmic-necessity claim. Event-locking of a measurement chart, if replicable, is a state-geometry constraint. Causal denoising remains unsupported.

A stronger anatomical implementation would identify thalamoreticular circuitry with the scalar variance-control term itself. Current evidence does not license that identification. The broader hypothesis considered here is that thalamocortical and thalamoreticular circuitry may contribute to state-dependent access control through gating, gain, dimensionality, diffusion geometry, or contextual routing — none of which is required to be scalar $\sigma(t)$. Independent physiology already treats thalamus as a modulator of functional cortical connectivity and distributed cognitive control, not merely a relay of categorical content (Halassa & Kastner, 2017; Schmitt et al., 2017). That is the external foundation for this access-control formulation. It is not identification of A_{noise}.

A companion two-operator specialization makes that distinction executable on a frozen landscape,

$$dX_t = A_{\text{sig}} b_0(X_t)dt + A_{\text{noise}}\Sigma_0 dW_t, \tag{19}$$

with V held fixed. Under that contract, directional A_{noise} can rotate short-horizon dispersion at matched total diffusion power, and non-normal A_{sig} can produce $\alpha < 0 < \omega$ while a scalar contraction cannot (Langell, 2026e). Those are **operator-existence** results on a declared 2-D potential, not thalamic physiology. Empirical TRN perturbation work currently reaches a lower rung: PV-selective photoinhibition is associated with evoked participation-ratio contraction, and both PV- and SST-family manipulations associate with non-scalar residual-covariance shape after prestimulus cleaning (Langell, 2026f; 2026g). That is a bounded contrast against scalar rescaling, not identification of A_{noise}. Closed gates remain in S3. A direct TRN $\to$ mesoscopic $\to$ surface-macro bridge is also not currently estimable from the audited public archives: the available meso–macro and TRN-perturbation resources do not contain all three links within a shared preparation, and cross-dataset pooling would not identify the missing causal chain. NDT has not found the thalamic mechanism.

9.2. History-dependent control

The landscape problem is independent of thalamic routing. Equation 9 can specify motion given V. It does not say where V came from. A history-conditioned landscape

$$V(x, t \mid H_t) \tag{20}$$

is the natural object for later transition probabilities, basin depth, and hysteresis. The Hippocampal–Entorhinal Landscape System (HELS) asks whether replay **writes** that landscape. Current infrastructure does not constitute evidence of writing. That HELS question remains speculative.

HELS-A is a different question. It asks whether a context-conditioned access operator can change which routes remain available **while V is held fixed**:

$$dX_t = -A_H \nabla V(X_t)dt + \sigma dW_t. \tag{21}$$

Companion simulations with known ground-truth dynamics support that split on declared geometries: directional A_H can saturate first-passage toward a context-matched target where speed-matched scalar gain does not, and an unsaturated committor can be biased by context and collapsed by shuffle. A neighbourhood robustness extension did not pass. The licensed statement is narrow: V fixed and X_0 fixed do not imply $q(X_0)$ fixed when A_H changes (Langell, 2026h). That is not a hippocampal, thalamic, memory, or MNPS result. HELS-A shows that the landscape problem and the access problem can come apart. The corresponding biological scoring gap is recorded in S3.

A separate observation — that true recent history can improve linear prediction of post-stimulation measured EEG beyond the current measured state on one meso-to-macro contract — is a P6-adjacent **measured-state** constraint. It is not landscape writing and not a scored hippocampal mechanism.

9.3. Physiological boundary conditions

Bodily signals are treated as context variables that may condition accessibility, not as coordinates of consciousness. Companion work can support bounded autonomic–chart co-modulation without identifying causal body-to-manifold gating. Any such coupling belongs in c_t^{body}, not in (m, d, e) by default.

Slower physiology is a feasibility boundary. Let $q^{\mathrm{met}(t)}$ denote metabolic adequacy. A trajectory may be geometrically available in the abstract model yet physiologically unavailable because the substrate cannot support it. Metabolism is not a consciousness scalar, not an MNPS axis, and not equivalent to arousal.

A slower dendritic mobility bias is a speculative microscale analogue of access without landscape rewrite; it is not an empirical MNPS identity (Langell, 2026i). Detail stays in S3.

10. Empirical constraints

No companion result validates the NDT chain end to end. The role of evidence here is to constrain layers, not to define the theory. Three anchors enter the main text because they answer different questions in the hierarchy of Table 1. Everything else is summarized as structured dissociation rather than narrated dataset by dataset. Every numerical claim below is bound to one canonical companion artifact, listed in S2. Superseded intermediate bundles are not used.

Companion simulations with known ground-truth dynamics, already used to bound a, α, and ω in Section 6 and Section 7, belong in this map as **methodological qualification**: they ask whether nominated mathematical objects recover known latent targets under inspectable envelopes. They do not identify simulator coordinates with MNPS and they do not replace empirical anchors.

10.1. Anchor 1: spindle-locked state reconfiguration

The first question is whether a physiologically discrete rhythmic event coincides with local chart reconfiguration — not whether spindles prove denoising, memory writing, or a thalamic variance clamp.

A companion overnight-EEG analysis of N2 spindles on a frozen stratified MNPS export — **Sleep Spindles Briefly Shift Cortical Network Geometry in Two Independent Overnight EEG Datasets** — finds a replicable within-event rise in the network-deviation coordinate d_n when the spindle window is compared with a far pre-event baseline (ANPHY high-density EEG, $N = 27$; BOAS / OpenNeuro ds005555, $N = 128$) (Langell, 2026j; López-Larraz et al., 2025; Wei et al., 2024). Effect sizes on that timing contrast are modest (ANPHY $+0.106$, $q = 0.009$; BOAS $+0.165$, $q = 0.011$). A third overnight polysomnography sample (RichSleep, $N = 24$) is underpowered for the same timing estimand and is treated as corroboration via its event-versus-matched-control contrast, not as an independent timing replication. Jacobian, diffusion-tensor, and reachability endpoints were not the primary objects. Canonical sources are listed in S2.

Matched-control specificity is incomplete. Several family endpoints are dominated by baseline differences rather than by the event increment, and the event-versus-matched-control sign for d_n conflicts across datasets for that reason. Autonomic composites do not moderate the geometry effect under the primary protocol. Sigma-band power residualization was not part of the frozen schema.

Anchor 1 ceiling

Discrete spindle events coincide with a replicable, timing-locked rise in stratified d_n. That is chart-level state reconfiguration at a rhythm-defined micro-event (claim class 1). It is not causal denoising, not a pure spindle-versus-matched-N2 mechanism, not a reachability change, and not TRN identification.

10.2. Anchor 2: recovery-linked geometry after cardiac-arrest coma

The second question is whether a clinically important state difference appears more clearly in accessible futures and local flow than in static location alone.

A companion pooled analysis — **Recovery-Linked Neural Geometry After Cardiac-Arrest Coma** — uses the final corrected 301-patient I-CARE dataset and evaluates MNPS spread, local Jacobian deformation, and short-horizon predictive-spread summaries jointly in two independently processed, non-overlapping cohorts (301 analyzable patients across 5322 recordings; 281 in the CPC 1 versus CPC 5 contrast, 141/140; window 0 –12 h after the EEG clock, not the ROSC clock) (Amorim et al., 2023; Langell, 2026k). All five prespecified predictive-spread / short-horizon reachability summaries survive BH-FDR in the pooled sample. The strongest individual effects are persistence/log-det volume proxies ($d = -0.909$ and $d = 0.896$); effective predictive-spread dimensionality is $d = 0.838$ pooled and already $d = 0.786$ in the first analyzable hour. Local Jacobian summaries are directionally aligned — lower expansion and deformation in later-good-outcome patients — but heterogeneous in magnitude across cohorts. Static MNPS spread is lower in better-outcome patients and is restored as a descriptive anchor, not as the lead. A subject-wise circular-shift null does not support a claim about which hour is uniquely strongest. Adding the geometry layer to a feature-limited clinical-plus-qEEG baseline raises internal AUROC from 0.599 to 0.662, a bounded increment that is not multimodal bedside prognostication. Canonical sources are listed in S2. An earlier ComBat-corrected 298-patient merge is not the source used here.

Because withdrawal-of-life-sustaining-therapy, time-to-death, and sedation annotations were unavailable, the contrast is recovery-linked geometry, not a pure viability assay and not a consciousness meter.

Anchor 2 ceiling

Early post-arrest EEG contains layered, recovery-linked geometry: short-horizon predictive-spread summaries and local deformation carry information that static severity/location does not exhaust. These are study-specific chart-level estimators. Residual quality and perturbation-response were not the study design, so the result motivates the accessible-futures layer. It does not constitute identification of a dynamical operator from known ground-truth dynamics.

10.3. Anchor 3: meso-to-macro contact without operator recovery

The third question is methodological. Can mesoscopic population quantities predict macroscopic EEG features under nulls, and does that contact license a local operator?

A companion analysis of simultaneous Neuropixels and surface EEG during cortical electrical stimulation, continuing from wakefulness into isoflurane (Claar et al., 2023; Langell, 2026b), reports median subject associations $r = 0.406$ (LFP SD with EEG SD) and $r = 0.395$ (spike rate with EEG spectral entropy), surviving sign-flip and circular time-shift nulls. Event alignment remains incompletely validated, so these are conditional trial-aligned associations. The same EEG-only local-linear chart yields a median residual-variance fraction of 0.856. Jacobian and reachability inference on that chart are therefore rejected.

A complementary IBL analysis reaches a related boundary: raw population decoding detects stimulus side under a pilot positive-control test, whereas a predefined single-trial chart-transition model fails its held-out estimability floor (Langell, 2026c). Failure of a dynamical readout need not imply absence of information in the underlying population.

The pair is the epistemic spine of the empirical map. Cross-scale contact is real at association. It is not common latent-state identity, unique NDT geometry, causal direction, TRN identification, or a validated MNJ. A visually plausible chart plus a correlation does not license a different inferential claim class.

Anchor 3 ceiling

Meso and macro observables can covary under stimulation-aligned windows after temporal and statistical nulls (claim class 1). The tested EEG chart does not recover a local flow, Jacobian, or reachability operator (claim class 4 fail).

10.4. Structured dissociations, not a universal signature

NDT predicts that different conditions move different layers. Companion clinical and task analyses are therefore kept as one constraint matrix rather than as additional anchors.

Setting	What moves	What does not	Ceiling
Depressive-symptom rest	reachability; 9D rotational structure	coarse 3D volume as lead	symptom-burden geometry, one young cohort
First-episode psychosis	medication-stratified chart and trajectory splits	one FEP signature	bounded; not a universal deformation
Gambling vs cognitive control	local Jacobian / flow geometry	MNPS location; matched EDA	state-versus-flow dissociation
Hypoglycaemic clamp	3D/9D state; slow-rhythm increase	confirmatory reachability collapse	feasibility constraint, not q^{met} identification
Infant age	chart coordinates and classical qEEG	FDR-supported local-flow change	development as state, not futures
Expected-stimulus omission (mouse VISp) (Langell, 2026l)	event-locked chart separation	Jacobian, diffusion, reachability, prediction-error ID	AVB-adapted chart; state geometry only
IBL chart readout (Langell, 2026c)	population decoder, stimulus-side pilot	single-trial discrete chart-transition operator	information can be present while the nominated dynamical readout fails
Human MTL–scalp (DANDI 000574) (Langell, 2026m)	neither primary frozen pair	meso–macro association under that contract	design-specific negative, not a replication of DANDI 000458

Table 3: Selected dissociations. NDT is constrained by which layer moves, not by a single consciousness-shaped manifold shift. Full companion reports remain outside this article.



Figure 5: Evidence map for the present theory. Cells are status under frozen contracts, not independent replication. V = validated; B = bounded; N = null or fail; dash = not tested; S = speculative. Added matrix rows: Allen AVB (state geometry), IBL chart readout (decoder succeeds / flow fails), and DANDI 000574 (human MTL–scalp association fail). Empty anesthesia and epilepsy rows are not tested, not hidden positives.

The pattern is the result. Reduced dynamical capacity is not itself reduced consciousness. Sleep, anesthesia, seizure, coma, and affective or psychotic regimes may all alter capacity-like summaries while differing in whether the change is mainly location, local deformation, stochastic spread, reachable volume, or canalized lock-in.

11. Neighbouring frameworks

NDT does not compete with constitutive or access theories of consciousness on their own terms. Its defensible claim is narrower: it is a framework for dynamical accessibility and a versioned measurement framework on which predictions from several of those theories can be expressed and, in places, cross-checked (Storm et al., 2024). The comparison that follows is therefore a map of partly different questions, not a contest for a single title. A longer thematic treatment of the same neighbourhood exists as a companion essay (Langell, 2026n); this section imports only the bridges and non-equivalences that constrain the present theory.

Positioning

IIT proposes what consciousness is claimed to be. GNWT proposes when content becomes globally available. NDT proposes how neural trajectories are locally generated, spread, gated, and made more or less reachable. Those are not interchangeable answers to one question.

The mathematical machinery — generators, Markov semigroups, transition-path theory and committors, transfer operators, and finite-horizon stochastic spread — is established (Deco et al., 2017; Klus et al., 2018; Metzner et al., 2009; Tognoli & Kelso, 2014). NDT's proposed novelty is not the discovery of a transition kernel. It is a neuroscientific synthesis: treating the geometry of a control-dependent future law as a primary object of neural-state organization; decomposing that law into non-equivalent empirical questions; separating measurement-chart claims from latent-process claims; testing whether context, history, or physiology changes the future law at matched chart state (C3); and applying that synthesis to consciousness-relevant regimes. Generators, diffusion tensors, committors, and related stochastic objects are established mathematics. The proposed contribution is their joint organization into present state, local law, future-law structure, and conditional access; the requirement that those objects not be treated as interchangeable; the measurement / latent-process separation; and the empirical commitments C2 and C3. The established stochastic-process machinery provides the mathematical scaffolding; the NDT claim lies in that decomposition, measurement discipline, and conditional-accessibility hypotheses.

11.1. Different questions, not rival scalars

"What does consciousness correspond to?" is not a single question. Neighbouring frameworks answer at least six different ones: whether experience is present (level); what is experienced (content); whether information is available to report and action (access); whether the system represents itself as being in a state (meta-awareness); whether experience is embodied; and what range of neural transitions remains available (dynamical capacity). NDT addresses the last of these. It does not currently identify NCC proper, and its charts are closest to enabling state dynamics — NCC prerequisites and the dynamical context around experience — rather than to report, confidence, or phenomenal identity.

Theories can therefore be nested as explanatory layers without being jointly validated (Storm et al., 2024). Predictive inference can supply candidate content (Clark, 2013; Friston, 2010). Recurrent processing can stabilize sensory representations (Lamme, 2006; 2010). A workspace can broadcast selected content (Baars, 1988; Dehaene & Changeux, 2011; Mashour et al., 2020). Higher-order or attention-schema systems can represent aspects of that access (Brown et al., 2019; Lau & Rosenthal, 2011; Webb & Graziano, 2015). Embodied inference can bias the background state (Seth, 2013). Dynamical geometry can describe the trajectory through which those processes unfold. Structural network control can constrain which transitions are cheap or costly (Gu et al., 2015; Tang & Bassett, 2018). Compatibility of layers is not evidence that every theory is true.

11.2. Closest family: landscapes, metastability, and control

NDT's nearest relatives are dynamical-landscape, metastability, and network-control accounts (Breakspear, 2017; Deco et al., 2011; 2017; Gu et al., 2015; Hancock et al., 2025; Kelso, 2012; Tognoli & Kelso, 2014). They already deny that regional activity magnitude is a sufficient description. A manifold visualization likewise does not identify the dynamical rule (Duncker & Sahani, 2021). NDT inherits that denial and adds five commitments that those families do not jointly require:

1. **Stochastic diffusion geometry** as a first-class local object $a(\xi, t)$, not only noise around an attractor.
2. **Accessibility decomposition**: location, local flow, stochastic spread, reachability, commitment, and finite-amplitude resilience answer different questions (Table 1).
3. **Explicit measurement contracts**: MNPS and NeuralManifoldDynamics are operational charts and estimators, not unique latent identification.
4. **State-dependent control**, of which rhythmic variance scheduling is the minimal special case, not the whole theory.
5. **Recoverability discipline**: association does not license a local operator (Section 3).

Network control theory shares the concern with possible transitions. NDT does not currently assume an explicit input matrix B or minimize control energy. Its reachability objects are empirical, chart-level, and short-horizon; they are not Gramian controllability of a structural connectome. Metastability shares the concern with a regime between lock-in and fragmentation. MNPS m is an operational mobility/coordination proxy, not a Kuramoto order parameter computed from pairwise phase relations. Landscape models share potential descent. In NDT that descent is a structured component of drift, not the general law: non-conservative flow, non-normal transients, and anisotropic diffusion remain in u_{nc} and a.

The claim "conscious states have geometry" is therefore not distinctive. The testable NDT increment is that neural variability and transition structure are systematically and state-dependently modulated, and that the resulting local flow, diffusion, and accessible futures can contain information not recoverable from location, power, or generic complexity alone.

11.3. Constitutive, access, and inference theories

Framework	Possible bridge	Non-equivalence
PP / FEP	precision; landscape descent; state estimation	fitted accessibility is not a generative model with explicit priors
IIT	both reject activity magnitude as sufficient	NDT does not identify consciousness with Φ or intrinsic causal structure
GNWT	access geometry may enable or disable global availability	dynamical reachability is not workspace ignition
RPT	rotational / local recurrent flow Ω	chart-level rotation is not anatomical feedback
HOT	none proposed	NDT does not model representation-of-representation
AST	attention-related coordinates as possible inputs	an attention proxy is not an attention schema
Landscapes / metastability	closest dynamical family	NDT adds diffusion geometry, control, and measurement contracts
Network control	possible transitions; future state space	no current B matrix or control-energy objective

Table 4: Bridges are analogies unless a shared formal object is identified. Superficial vocabulary overlap is not equivalence.

Predictive processing / free-energy principle. Precision, inference, and embodied prediction supply a neighbouring language for content and for slow bodily context (Clark, 2013; Friston, 2010; Seth, 2013). The relevant comparison is a specified generative model, not the umbrella principle. NDT's emphasis is measured, state-dependent transition structure. A fitted drift term is not variational free energy, and $r(t)$ is not precision by another name.

Integrated information. IIT identifies consciousness with irreducible intrinsic cause–effect structure, quantified as Φ (Oizumi et al., 2014; Tononi et al., 2016). Perturbational complexity is clinically important and IIT-inspired, but it is not an empirical calculation of Φ (Casali et al., 2013). NDT refuses a structural "amount of consciousness" scalar of that kind. Shared rejection of simple magnitude is not a shared ontology.

Global workspace. GNWT targets access: information becomes globally available after a competitive ignition-like transition (Baars, 1988; Dehaene & Changeux, 2011; Mashour et al., 2020). Dynamical accessibility in NDT may describe conditions that make such a transition more or less available. It does not model broadcast, report, or workspace competition. Reachable volume is not ignition.

Recurrent processing. RPT locates a threshold in recurrent sensory processing rather than in global broadcast (Lamme, 2006; 2010). NDT's antisymmetric Jacobian Ω is compatible with rotational or recurrent flow. Compatibility is not anatomy: a chart rotation does not establish feedback in sensory cortex, and recurrence is not treated as sufficient for experience.

Higher-order theories and attention schema theory. These target meta-representation and a simplified internal model of attention (Brown et al., 2019; Lau & Rosenthal, 2011; Webb & Graziano, 2015). NDT currently has no corresponding state variable. Metacognitive sensitivity and awareness attribution are neighbouring empirical programmes, not NDT objects.

11.4. Lessons that apply across theories

Two external results discipline every framework in this neighbourhood, including NDT.

The Cogitate Consortium's preregistered adversarial collaboration did not declare a winner between IIT and GNWT (Cogitate Consortium et al., 2025). Predicted posterior synchronization and predicted prefrontal offset ignition were both challenged in the specified form. The methodological lesson is prior commitment: mature theories become stronger when they state in advance what would count against them. NDT has not undergone an adversarial test of comparable independence or scale.

State discrimination is not detection of experience. Complexity measures can separate wakefulness from sedation while failing to distinguish reported dreaming from no reported experience within the sedated condition (Bajwa et al., 2025). PCI, ignition markers, and NDT reachability may all index regimes that make experience more or less likely. None is licensed as a consciousness meter by that fact. Companion coma geometry is recovery-linked capacity (Section 10), not a within-state experience assay. Dynamic patterns of brain coordination

also differ across conscious and unconscious conditions, including disorders of consciousness (Demertzi et al., 2019). Content or representational geometry — for example no-report colour similarity — is a neighbouring question, not an accessibility measurement.

12. Predictions and failure conditions

NDT is tested by layer, not as a single slogan. The predictions below are theory-level. Companion anchors constrain some of them; none is a completed adversarial result.

Prediction	What would fail it	Present constraint
P1 Rhythmic control	event structure adds nothing beyond power, autocorrelation, smooth precision, or event density	spindle-locked d_n timing; schedule specificity incomplete
P2 State vs accessibility	richer dynamics add no replicable information beyond chart location	coma, gambling, and depression move different layers
P3 Linear vs finite-amplitude	FAR collapses onto validated local-linear summaries	conceptual; not an empirical NMD export
P4 Diffusion vs drift	anisotropy / dimensionality add nothing once drift is controlled	simulator envelopes only
P5 Thalamic access	thalamic variables add nothing beyond arousal / sensory gain	association and synthetic operators; not biology ID
P6 History dependence	$P(Y_{t+h} \mid x_t, H_t) \approx P(Y_{t+h} \mid x_t)$	measured-state prediction constraint; not landscape writing
P7 Embodied conditioning	body-state effects vanish under autonomic / arousal controls	timing support; spindle-geometry efficacy rejected
P8 Within-state experience	after matching state, responsiveness, arousal, power, and complexity, K_h^x still fails to split reported experience from no reported experience	not tested; Bajwa-type within-state designs are the template

Table 5: Falsifiable predictions. Success or failure of one row revises that layer. It does not automatically ratify or refute the modelling class. P8 is the within-state experience test; it has not yet been tested.

A stronger NDT discriminant would jointly require event specificity, schedule specificity against stationary and smooth controls, incremental value beyond power and generic complexity, and failure under phase-shuffle, time-shift, wrong-event, and power-matched nulls. Current spindle evidence supports a timing-locked chart change. It does not yet satisfy the full discriminant.

P8 is the headline within-state test if the article is read as a consciousness paper. After matching global physiological state, responsiveness, arousal, spectral power, and standard complexity, does

$$K_h^x(\cdot \mid x_t, \text{experience}) \neq K_h^x(\cdot \mid x_t, \text{no experience}) \tag{22}$$

(Bajwa et al., 2025)? A positive result would license a stronger claim about mind-relevant accessibility. A negative result would constrain any stronger claim about mind-relevant accessibility. The present manuscript does not promote P8 as already answered by coma geometry or complexity scores (Table 1).

Hierarchical failure

Falsification of one subsystem does not falsify dynamical accessibility as a scientific object. TRN failure would drop a biological implementation. RVC failure would strongly constrain rhythmic scheduling. MNPS failure would retire that chart. Operator-recovery failure on a given recording, already observed on DANDI 000458, retires that estimator on that chart. None of those results implies that neural states have no structured futures.

NDT should be substantially narrowed if, across reasonable alternative charts and nulls: rhythmic scheduling adds no held-out value; local-dynamics estimators repeatedly fail ground-truth or residual-quality tests; diffusion, reachability, or FAR reduce to redundant versions of simpler statistics; history adds no conditional

information; and apparent clinical or altered-state effects are exhausted by power, movement, arousal, or preprocessing artifacts. The constitutive reading that rhythmic reconstruction **is** experience remains a labeled speculative extension. It currently adds no discriminating empirical prediction beyond the dynamical model.

13. Discussion

13.1. Dynamical accessibility as the organizing principle

NDT treats neural-state location, local dynamics, possible futures, and their control as distinct but coupled levels of description. Rhythmic variance scheduling provides one experimentally specific control hypothesis; learned landscapes provide one structured component of drift; MNPS provides one measurement chart; and diffusion, Jacobian dynamics, reachability, committors, and finite-amplitude resilience interrogate different aspects of the conditional future law. None is individually synonymous with NDT. This formulation subsumes the earlier rhythmically scheduled denoising model as a restricted case in which control acts primarily through scalar stochastic variance (Langell, 2026a).

Companion constraints illustrate the decomposition rather than a single signature: spindle events reconfigure a stratified chart without licensing causal denoising; early post-arrest EEG separates more clearly on reachable futures and local deformation than on static spread; meso-to-macro contact does not recover a local operator (Section 10). Charts and estimators can fail without making dynamical accessibility meaningless, and they can succeed as measurements without identifying the latent law. A consciousness scalar, a universal clinical fingerprint, and any reading in which MNPS location **is** the NDT claim remain outside the theory.

13.2. Why this may matter

A conscious brain state may not be distinguished purely by complexity, entropy, synchrony, occupancy, or global activity. Candidate regimes may also differ in **what the system remains capable of becoming next**. That is a hypothesis about the organization of neural states associated with wakefulness, sleep, anesthesia, task engagement, and pathology. It is not an identity claim about experience, not a meter of awareness, and not a solution to the hard problem.

The hypothesis is useful only if the layers dissociate. If reachability, local flow, and diffusion were redundant encodings of location or power, the accessibility hypothesis would collapse. The present empirical map is not a validation of the chain. It is evidence that the collapse has not already occurred in every tested regime, and that different conditions move different layers (Table 3). Consciousness research already contains neighbouring theories of content, access, recurrence, and intrinsic causal structure (Section 11). NDT's contribution, if it holds, is a measurement surface for dynamical capacity and its control — a surface on which those theories can make distinct predictions, not a replacement for them.

The next informative experiment is therefore not another broad disease contrast. It is a design that separates global state, responsiveness, reportability, and momentary experience, and that asks whether rhythm-locked or context-locked changes in local flow, diffusion, and short-horizon reachability explain future trajectories beyond power, generic complexity, and stationary dynamics (Section 12). A positive result would still support a neural mechanism, not the claim that reconstruction is consciousness.

14. Limitations

No dataset currently validates the NDT chain end to end. Many nominated objects remain estimator-sensitive: MNPS is operational and chart-dependent; drift and diffusion are identifiable only under explicit assumptions; local linearization fails in important empirical regimes, including the EEG chart tested on DANDI 000458; reachability depends on horizon and noise model; committor estimation requires meaningful destinations that current measurement contracts do not emit; and finite-amplitude resilience depends on perturbation family and amplitude scale.

Biological bridges remain uneven. Rhythmic variance control is a distinctive prediction, not a demonstrated executor. Thalamic access has synthetic operator existence and associative biology, not circuit identification. History-dependent landscape writing is open; access on a frozen landscape is a computational existence result, not scored hippocampal mechanism. Embodiment and metabolism are contextual and feasibility constraints, not consciousness coordinates. Dendritic slow-branch accessibility is a simulator discriminator and a speculative micro substrate.

Empirical companions are bounded. Sleep, coma, and meso-to-macro analyses constrain selected layers in selected cohorts. Clinical fingerprints are not biomarkers. Treatment-limitation and sedation confounds remain unresolved where they were unavailable. NDT has not undergone an independent adversarial test comparable

to Cogitate. Apparent geometric effects must still survive alternative charts, power-matched and temporally shuffled nulls, and standard spectral, connectivity, and complexity baselines.

These limits are scope, not temporary embarrassment. NDT is a framework for neural dynamical organization associated with conscious-state regimes. It does not explain why those dynamics are accompanied by experience.

15. Conclusion

Noetic Diffusion Theory proposes that neural-state organization is incompletely described by state position alone. A neural state also carries a local dynamical structure: deterministic tendencies, stochastic directions, transient sensitivities, reachable futures, basin commitments, and finite-amplitude resilience. Those structures are themselves modulated by rhythm, context, history, physiology, and candidate access-control systems. MNPS supplies one measurable chart through which this organization can be studied. NeuralManifoldDynamics supplies one implementation rather than the definition of the theory. Rhythmic Variance Control supplies the minimal control case rather than the whole mechanism.

The resulting framework predicts structured dissociations among location, stability, diffusion, accessibility, commitment, and resilience, and treats those dissociations as empirically testable correlates of changing neural regimes — not as a scalar of consciousness, and not as a complete explanation of experience. What a neural state **is**, scientifically, includes what it can still become.

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Companion analyses cited in the empirical section are archived on Zenodo. Source datasets used there are cited with the corresponding anchors: ANPHY and BOAS/ds005555 (spindles), I-CARE (coma), and DANDI 000458 (meso–macro). This theoretical article does not release a new recording.

Competing Interests

The author declares no competing interests.